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Enhancing methane production of anaerobic digested sewage sludge by advanced wet oxidation & steam explosion pretreatment

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ABSTRACT

Anaerobic digestion (AD) of sewage sludge is associated with lowered carbon conversion efficiency due to the recalcitrance of microbial cells in the Waste Activated Sludge (WAS) from the aeration basins with 50 % of the dry weight still being present after AD. The effect of Advanced Wet Oxidation & Steam Explosion (AWOEx) pretreatment on the methane yield at variable temperatures (165–200 °C), residence time (15–45 min) and oxygen dosage (1%–10% based on VS concentration) was tested in this study. Central composite design (JMP® Pro) was used to design 19 runs of AWOEx pretreatment of dewatered digested sewage sludge (DADSS) at different operating conditions. After pretreatment the DADSS was fed to 19 bioreactors operated in semi-continues mode at 37 °C. The highest average methane yields of 183 mL/g VS and 170 mL/g VS were found for the bioreactors receiving DADSS pretreated at 165 °C with a retention time of 15 min and 10 % O_2 (Condition 1), and, 182.5 °C with a retention time of 15 min and 5.5 % O_2 (Condition 2). This corresponds to an increase in the methane production of 156.2 % and 140.5 % compared to the methane production from the DADSS without pretreatment.

Plotting the results using the response surface model showed that temperature and O_2 dosage had a significant effect on the methane yield while residence time was of lower importance. Importantly, thermic treatment without oxygen showed far lower methane yields of 114.5 mL/gVS and 120.26 mL/gVS, respectively of the two best conditions described above. Through this two-step AD process, we were able to achieve a net 94% improved methane yield compared to conventional AD of sewage sludge by implementing a second AWOEx pretreatment step (followed by secondary AD) with 26.9% decreased organics content, further reducing the cost of the final disposal of the treated sewage sludge.

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1. Introduction

In United States a total of 6.5 million metric tons per year of dewatered sewage sludge (DSS) is produced in the ca. 15,000 wastewater treatment plants (WWTPs) (Nguyen et al., 2021; Seiple et al., 2020). The high organic content of DSS

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makes this waste a potential resource for production of bioenergy and biofuels (Shi et al., 2021). Approximately 50% of all WWTP's in the US corresponding to a total number of 7500 are converting their sewage sludge to biogas (a mixture of methane (ca. 60%) and CO_2 (ca. 40%)) using anaerobic digestion (AD) (Shi et al., 2021). However, the amount of biogas produced are often too low to warrant the use of this biogas. If more biogas could be produced from the sewage sludge this could offset a significant amount of the energy used by WWTPs. Besides, biogas can be upgraded to Renewable Natural Gas (RNG) and used as a transportation fuel or added to the natural gas grid. A major constraint for producing biogas as an energy source is the fact that only approximately 50% of the organic material in the sewage sludge will be converted during conventional AD. It has previously been reported that conventional AD of sewage sludge produce a methane yield in the range of 0.19 to 0.24 $Nm^3/Kg\ VS_{in}$ (Di Capua et al., 2020). The residual organics after AD needs to be concentrated by separation and disposed either in landfills, on agricultural land and parks or incinerated (Nguyen et al., 2021). All these management practices have some specific problems for both the environment and the public.

Typically, the composition of concentrated sewage sludge includes TS (12%–40%), VS (75%–85%), protein (18%–40%), lipids (6%–19%), lignin (23%–29%), and carbohydrates (10%–15%) (Nguyen et al., 2021; Su et al., 2019). One of the major reasons for the recalcitrance of sewage sludge during AD is the fact that activated sludge (WAS) is mainly composed of bacterial cell walls as well as recalcitrant polymers such as cellulose, hemicellulose and lignin (Hao et al., 2020; Nguyen et al., 2021). Previous studies have shown that the digestate produced after AD is still rich in organics, which have not been converted during the AD process (Li et al., 2017; Nordell et al., 2021; Wu et al., 2020). Several thermal pretreatments technologies have been developed for enhancing AD of dewatered sewage sludge (Cambi THP, Veolia Exelys) (Abu-Orf and Goss, 2012; Shi et al., 2021). However, none of these techniques has been applied to DADSS as tested in this study. The conditions used for both these processes are generally less severe (temperatures of max 160 °C) and the biogas yields improvement are generally moderate around 30%–40% (Nguyen et al., 2021; Seiple et al., 2020; Wang et al., 2021; Yan et al., 2021).

Advanced Wet Oxidation & Steam Explosion pretreatment (AWOEx) is a thermochemical process integrating wet oxidation and steam explosion (Ahring and Munck, 2013; Miklos et al., 2018; Svensson et al., 2018; Wu et al., 2020). In this process, biomass is exposed to an oxidizing agents such as air, hydrogen peroxide, pure oxygen etc. under high temperatures (over 140 °C) and pressures (over 10 bars) for the duration of 15 to 45 min (Ahring and Munck, 2013; Biswas et al., 2015, 2020; M'Arimi et al., 2020). AWOEx pretreatment of lignocellulosic biomass materials significantly enhance lignin solubilization. In addition, crystalline cellulose remains in the solid fraction whereas most hemicellulose components are solubilized during the process (Biswas et al., 2020; Khan and Ahring, 2020). As a result of AWOEx pretreatment, the surface area of the biomass is greatly increased, which increases the microbial and enzymatic access to the carbohydrates in the material (Dewil et al., 2017). Besides, acetic acid is produced via de-esterification of the acetyl groups, which further influences the dissolution of hemicellulose into the aqueous phase (Biswas et al., 2020, 2014). The advantages of AWOEx pretreatment are besides its effectiveness the fact that no chemicals are added needing recovery along with a low formation of inhibitory by-products. AWOEx pretreatment can be performed at high dry matter concentrations of up to 45%, which lower the overall cost of the pretreatment (Biswas et al., 2020).

While AWOEx has been extensively tested for different lignocellulosic biomass materials (Biswas et al., 2015, 2014; Khan and Ahring, 2020; M'Arimi et al., 2020), the use of the technology for improving the carbon conversion efficiency of sewage sludge into methane has not been studied in any details. In this study, we will use a central composite statistic method to determine the optimal pretreatment conditions for AWOEx. We will further determine the composition of dewatered digested sewage sludge (DADSS) out of the initial AD bioreactor and out of the secondary AD to understand the effect that the AWOEx pretreatment would have on the biodegradation of the different components of digested sewage sludge.

2. Materials and methods

2.1. Feedstock and composition

Dewatered anaerobic digested sewage sludge (DADSS) was obtained from Walla Walla wastewater treatment plant in Eastern Washington. The DADSS was collected after dewatering in a filter press. It contained a total solids (TS) of 13.6%; a moisture content (MC) of 86.4%; and a volatile solid (VS) of 10.6%. All samples of DADSS before and after pretreatment were stored at 4 °C before use.

2.2. Design of experiment

The Advanced Wet Oxidation & Steam Explosion (AWOEx) pretreatment of DADSS was performed at temperatures ranging from 165–200 °C, oxygen (O_2) dosage from 1%–10% of the VS, and a residence time of between 15 and 45 min. A central composite design based on statistical analytic system (SAS) (Heng et al., 2021) was utilized to find the optimal pretreatment process using different dosages of oxygen (between 0%–10%) at temperatures ranging from 165 to 200 °C in a pilot pretreatment reactor (Supaporn and Yeom, 2022). From the model, we defined three AWOEx sets at the temperatures: 165 °C, 182.5 °C and 200 °C with and without oxygen added and with controls without any pretreatment (Table 1). The results from a series of 19 runs (3 controls and 16 test sets) with different operating conditions was used to generate the regression report of the factors and results.

Table 1
Characteristics of dewatered anaerobic digested sewage sludge (DADSS).

Items	Values
Moisture (%)	86.44
Total solids (%)	13.6
Volatile solids (%)	10.6
Ash (%)	2.92
Cellulose (%)	2.3
Hemicellulose	2.9
Lignin	0.9
Crude fat (%)	1.69
Protein (%)	3.01
Calories (Kcal/100 g)	15

2.3. Advanced wet oxidation & steam explosion pretreatment

AWOEx of DADSS was carried out in a 100 L stainless steel pressure tank. In each run 10 kg dewatered DADSS was feed into the reactor and kept at a mixing speed of 50 rpm. The target temperature was achieved by steam addition into the pretreatment reactor while the temperature was kept constant using an external oil heater. Oxygen (O_2) was purged into the reactor based on the amount of VS of DADSS while increasing the mixing speed to 85 rpm. Upon completion of the residence time, the material was flashed out of the tank and into a 250 L flash tank. After cooling, the sample was collected and stored at 4 °C for further analysis and biogas yield determination.

2.4. Anaerobic digestion of pretreated DADSS

Anaerobic digestion (AD) of DADSS were conducted in a series of 1 L bioreactors, operated in semi-continuous mode with a working volume of 400 mL. The hydraulic retention time (HRT) was kept at 20 days and all bioreactors were feed every second day at the exact same time with a recording of the biogas produced. Two reactors containing DADSS without pretreatment was used as controls.

2.5. Composition analysis

The moisture contents (MC), total solids (TS), volatile solids (VS), carbohydrates (cellulose and hemicellulose), lignin, and lipid content of all DADSS samples before and after AWOEx and AD were measured following the National Renewable Energy Laboratory (NREL) methods as previous described studies (Biswas et al., 2020; Hames et al., 2008; Van Wychen et al., 2016). Protein contents of all samples before and after AWOEx of DADSS were measured by the Lowry method (Amin et al., 2021; Judickaite et al., 2021). NREL standard method was used for determining total lignin (acid soluble and acid insoluble) and total cellulose and hemicellulose (Sluiter et al., 2008). Acid soluble lignin was determined using a UV spectrophotometer, whereas total carbohydrates (sum of glucose, xylose, and arabinose) were determined after two-step hydrolysis (Jenway 6405 UV/Visible, NJ, USA) with a wavelength of 320 nm. Monomers were analyzed by using an UltiMate® 3000 HPLC system (Dionex, Sunnyvale, CA) with an Aminex® 87H Column 250 × 4.6 mm (Bio-Rad, Hercules, CA) and a Shodex RI-101 refractive index detector at 60 °C using 4 mmol H_2SO_4 as an eluent with a flow rate of 0.6 ml/min. All measurements of lignin and carbohydrates analysis were done in duplicates. The lipid extraction was performed following chloroform–methanol extraction method proposed by Bligh and Dyer (1959).1 g of dried sludge were mixed with 20 mL of mixed liquor (chloroform: methanol = 2:1, v/v) and then subjected to 60 °C for 4 h. Thereafter, the mixture was placed in a centrifuge for 15 min at 3000 rpm and the solvent phase was transferred to a glass vial. After evaporating the solvent at 60 °C, the vial was weighed. The difference in weight was the total lipid content.

2.6. Volatile fatty acids analysis

The concentration of volatile fatty acids was determined in the influent and effluent samples for each bioreactor using high performance liquid chromatography (HPLC) as previously described (Khan and Ahring, 2020). 2 mL of each sample was taken and centrifuged at 10, 000 RPM for 10 min. The supernatant was separated with pipette, diluted 2 times with 4 mM sulfuric acid and the filtered through 0.2 μm. The samples were analyzed by UltiMate® 3000 HPLC system (Dionex, Sunnyvale, CA) with an Aminex® 87H column 250 × 4.6 mm (Bio-Rad, Hercules, CA) and a Shodex RI-101 refractive index detector.

2.7. Biogas analysis

The composition of methane and carbon dioxide from all bioreactors were determined using an Universal gas analyzer, UGA series (Stanford Research Systems, Sunnyvale, CA) (Khan and Ahring, 2020).

Table 2
AWOEx conditions for dewatered anaerobic digested sewage sludge.

Run	Temperature (°C)	Residence time (min)	O_2 dosage
1	200	30	5.5
2	182.5	45	5.5
3	182.5	30	10
4	165	15	1
5	182.5	15	5.5
6	200	45	1
7	200	45	10
8	165	45	1
9	182.5	30	5.5
10	165	30	5.5
11	182.5	30	1
12	200	15	10
13	182.5	30	5.5
14	165	45	10
15	165	15	10
16	200	15	1
17	182.5	30	0
18	165	30	0
19	200	30	0

Table 3
ANOVA test analysis on the effect of pre-treatment on DADSS.

	SS	DF	MS	P Value
Pretreatment condition (between groups)	606567	18	33698	$P < 0.05$
Pretreatment condition (within a specific group)	201036	570	352	$P < 0.05$

2.8. Statistical analysis

The design of the experiments and the experimental data were analyzed using the statistical software (SAS) JMP Pro (version 15.2.0) (Biswas et al., 2020). The response of temperature, residence time, and oxygen dosage on DADSS was evaluated using the methane yield obtained from the AD reactors operated under different conditions. The analysis was conducted on a fitted correlation using 2nd degree interaction under response surface macros and standard least-square personality with an emphasis on effect leverage (Biswas et al., 2015). The effect of pretreatment on the methane yield was analyzed through one way ANOVA test.

3. Results and discussion

3.1. Composition of DADSS

The characteristics of DADSS are shown in Table 1. DADSS contained 86.44% of moisture contents (MS), 13.56% of total solids (TS), and 10.6% of volatile solids (VS) which is 77.8% of TS. The results were similar to previously reported data (Nordell et al., 2021; Usman et al., 2019). Further, it was found that DADSS contained 2.3% cellulose, 2.9% hemicellulose, 0.9% lignin, 1.69% of crude fat and 3.01% of protein.

3.2. Effect of advanced wet oxidation pretreatment on DADSS

The optimization of AWOEx pretreatment of DADSS for high methane yield was based on the statistical experiment design of 19 pretreatment runs (Table 2), conducted in the pretreatment reactor with an initial concentration of 13.5% of total solids. The operating parameters: temperature between 165 and 200 °C, residence time between 15and 45 min and oxygen dosage between 1 and 10% of the VS of the samples tested, were used as the factors tested in the design of the experiments. To access the oxidation effect, three runs (17, 18 and 19) were performed without oxygen at a temperature of 165 °C, 182.5 °C and 200 °C at 30 min, and two runs (9 and 13) at central points of the operating parameters were conducted at temperature of 182.5 °C, 30 min retention time and O_2 of 5.5% to access the process variability. Again, upon analysis it was seen that the pre-treatment had significant effect on the methane yield across various experimental groups as the *P*-value was found to be lower than 0.5. This suggests that, our findings reject null hypothesis and supports alternative hypothesis since there is significant difference between the means of the experimental groups (Table 3).

3.2.1. Effect of AWOEx pretreatment on the composition of DADSS

The composition of DADSS after AWOEx pretreatment at different process conditions were determined to evaluate the effects of process parameters on the composition of DADSS. It was found that the protein concentration (4.90%), carbohydrates (2.85%) and lipid (1.01%) contents were higher at 165 °C compared to protein (4.57%), carbohydrates (2.70%)

Table 4
Composition of AWOEx pretreated samples of DADSS at different process conditions.

Sr #	Pretreatment operating conditions	MC (%)	TS (%)	VS (%)	Ash (%)	Protein (%)	Crude fat (%)	Carbohydrates (%)
1	200 °C; 30 min; 5.5%	91.82	8.17	76.95	2.13	4.47	< 1.3	2.62
2	182.5 °C; 45 min; 5.5%	90.99	9.01	75.99	2.21	4.52	< 1.3	2.75
3	182.5 °C; 30 min; 10%	90.78	9.21	76.56	2.24	4.64	< 1.3	2.68
4	165 °C; 15 min; 1%	91.47	8.52	77.34	2.32	5.42	< 1.3	2.86
5	182.5 °C; 15 min; 5.5%	91.47	8.51	78.97	2.20	4.68	< 1.3	2.81
6	200 °C; 45 min; 1%	91.54	8.45	75.17	2.18	4.46	< 1.3	2.65
7	200 °C; 45 min; 10%	91.50	8.49	75.49	2.15	4.43	< 1.3	2.60
8	165 °C; 45 min; 1%	91.64	8.35	77.39	2.36	4.92	< 1.3	2.84
9	182.5 °C; 30 min; 5.5%	91.76	8.23	77.24	2.27	4.60	< 1.3	2.71
10	165 °C; 30 min; 5.5%	90.81	9.18	78.17	2.48	4.83	< 1.3	2.92
11	182.5 °C; 30 min; 1%	90.43	9.56	76.46	2.25	4.63	< 1.3	2.80
12	200 °C; 15 min; 10%	90.76	9.23	75.93	2.14	4.38	< 1.3	2.63
13	182.5 °C; 30 min; 5.5%	91.87	8.12	76.38	2.26	4.51	< 1.3	2.71
14	165 °C; 45 min; 10%	91.07	8.92	77.01	2.39	4.95	< 1.3	2.98
15	165 °C; 15 min; 10%	90.35	9.64	77.38	2.36	5.41	< 1.3	2.95
16	200 °C; 15 min; 1%	91.77	8.22	76.79	2.12	4.45	< 1.3	2.57
17	182.5 °C; 30 min; 0%	90.56	9.43	77.14	2.28	4.57	< 1.3	2.70
18	165 °C; 30 min; 0%	90.94	9.06	77.65	2.80	4.90	< 1.3	2.85
19	200 °C; 30 min; 0%	91.14	8.86	75.75	2.15	4.42	< 1.3	2.55

and lipid (0.83%) at 182.5 °C, and protein (4.42%), carbohydrates (2.55%) and lipids (0.81%) at 200 °C for 30 min without oxygen. This can be explained by a higher hydrolysis at higher temperatures in accordance with the fact that the VS content was decreased when the temperature was increased in the pre-treated sets.

When oxygen 1 to 10% was added based on VS during the AWOEx pretreatment of DADSS it was found that the protein, carbohydrates, and lipid contents vary between 4.83–5.42%, 2.84–2.98 0.63–1.20%, respectively, at 165 °C with a retention time between 15 and 45 min. At 182.5 °C at the same range of retention times, protein, carbohydrates, and lipids contents were found in the range of 4.51–4.68%, 2.68–2.81% and 0.7–0.9%, respectively. At 200 °C at the same range of retention times, protein, carbohydrates, and ash contents were found in the range of 4.38–4.47%, 2.57–2.65% and 2.13–2.18%, respectively. It should be noticed that all operating parameters (temperature, residence time and oxygen dosage) affected the dissolution of DADSS during AWOEx pretreatment, resulting in larger portion of insoluble organics were converted into soluble organics at the highest setting for each parameter (Table 4).

3.3. Effect of advance wet oxidation & steam explosion pretreatment on methane production potential from DADSS

Fig. 1 shows the biogas production potential from AWOEx pretreatment of DADSS material at 19 different operational conditions while Fig. 2 shows the trend in methane yields from the 2 optimally performing semi-continuous bioreactors at 182.5 °C,15 min residence time with 5.5% O_2 and 165 °C, 15 min and 10% O_2 for 3 retention times with respect to AD1(control). The control reactor was feed with DADSS without AWOEx pretreatment to compare the effect of the different conditions during AWOEx pretreatment on methane production. It was found that the average methane yield of the control reactor was 72.33 mL/gVS. Further, it was found that thermic pretreatment of DADSS at 165 °C and 182.5 °C without oxygen dosage with 30 min retention time, the methane yields of the bioreactors were 123.66 and 126 mL/gVS, respectively. The maximum methane yields of 174.9 and 186.5 mL/gVS were obtained at 182.5 °C and 165 °C with oxygen added at a level of 5.5 and 10%. Overall, it was found that AWOEx pretreatment with oxygen dosage significantly affected AD of DADSS digestion, resulting in the improvement of average methane yield by 37.9 and 46.4% at 165 °C and 182.5 °C, respectively compared to the control sets pretreated with identical temperatures but no oxygen added. The average methane yields achieved under optimal condition 1 was 183.4 $\pm$ 2.4 mL/gVS at the 3rd retention time. For the bioreactor set under optimal condition 2, the average methane yields reported for the 3rd retention time was 170.1 $\pm$ 4.1 mL/gVS. Since the residence time was nearly constant after stabilization, it can be concluded that both temperature and O_2 % dosage during pretreatment significantly affects the methane yield of the materials. Interestingly, the highest methane yields of 186.5 mL/gVS under optimal condition 1 and 174.8 mL/gVS under optimal condition 2 was for the same residence time but with different temperatures and oxygen dosages, indicating that lower temperature can promote the same hydrolysis degree when the oxygen dosage is increased and also indicate that higher temperatures can reduce the need for oxygen addition. The results for the control reactor is in accordance with the study by Nordell et al. (2021), which found that only 46 mL/g VS of methane yield was achieved during the anaerobic digestion of digestate under thermophilic condition without any thermochemical pretreatment. Overall, AWOEx pretreatment of DADSS with oxygen significantly increased the methane yield corresponding to the result of the control reactor. Upon analyzing the difference of TS and VS % of all bioreactors, it was found that the two bioreactors with the highest methane yields under condition 1 and 2 also showed the highest TS and VS conversion of 43.92% and 43.15% under the optimum conditions prevailing at 165 °C (condition 1) and 182.5 °C (condition 2), respectively. This was further the highest TS/VS ratio of 1.41 and 1.39, respectively compared to the other bioreactors. Pearson correlation was performed to determine the linear correlation between methane yield

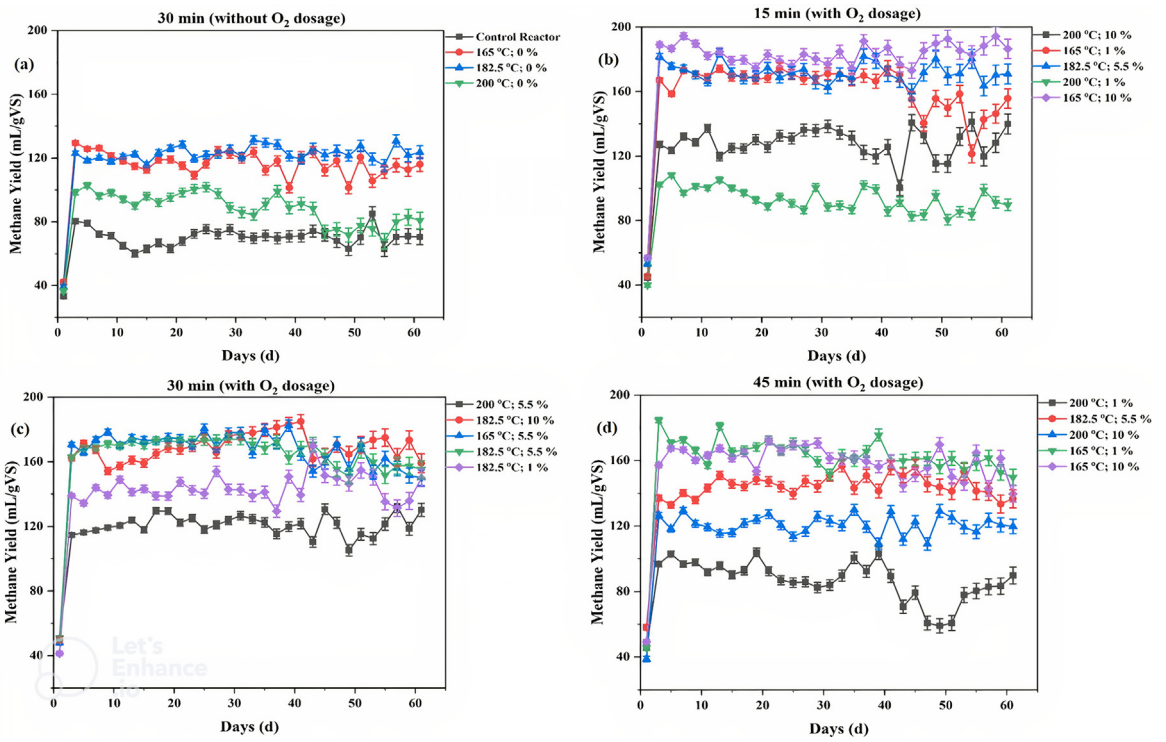


Fig. 1. Biogas production potential from AWOEx pretreatment of DADSS material at different operating conditions analyzed by the SAS software.

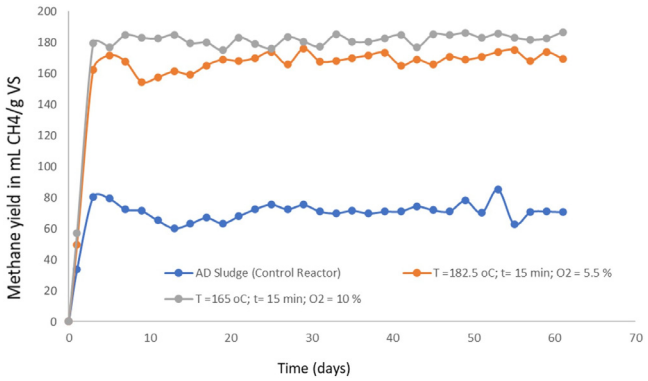


Fig. 2. Methane yield in the bioreactors at the optimal pretreatment conditions compared to control (no pretreatment).

and TS/VS ratio by the SPSS software (version 20.0) (Usman et al., 2020; Xiao et al., 2016; Yang et al., 2019). Pearson's correlation coefficient of 0.968 was found (+1, strong correlation; −1, weak correlation and 0, no relationship), indicated a strong relationship between increasing methane yields and TS/VS ratios (Usman et al., 2020).

3.4. Influence of the different operational conditions of AWOEx pretreatment of DADSS on methane yield and response surface modeling

Fig. 3 shows the methane yield after the AWOEx pretreatment of DADSS at different process conditions. It was found that all process parameters (temperature, residence time and O_2 dosage) had influence on the methane yield. Temperature and O_2 dosage had, however, significant effect on methane yield regardless of the residence time as indicated by high LogWorth (LogWorth > 2, significant) (Table 5). For AWOEx pretreatment, the temperature is a key operating parameter. The impact of pretreatment temperature on digester performance has been examined in various laboratory scale studies (Li and Noike, 1992). Earlier studies showed that gas production yields increases steadily with temperature from 120 to 175 °C (Biswas et al., 2020; Higgins et al., 2017). Similarly, Haug et al. (1978) found that gas production increased with

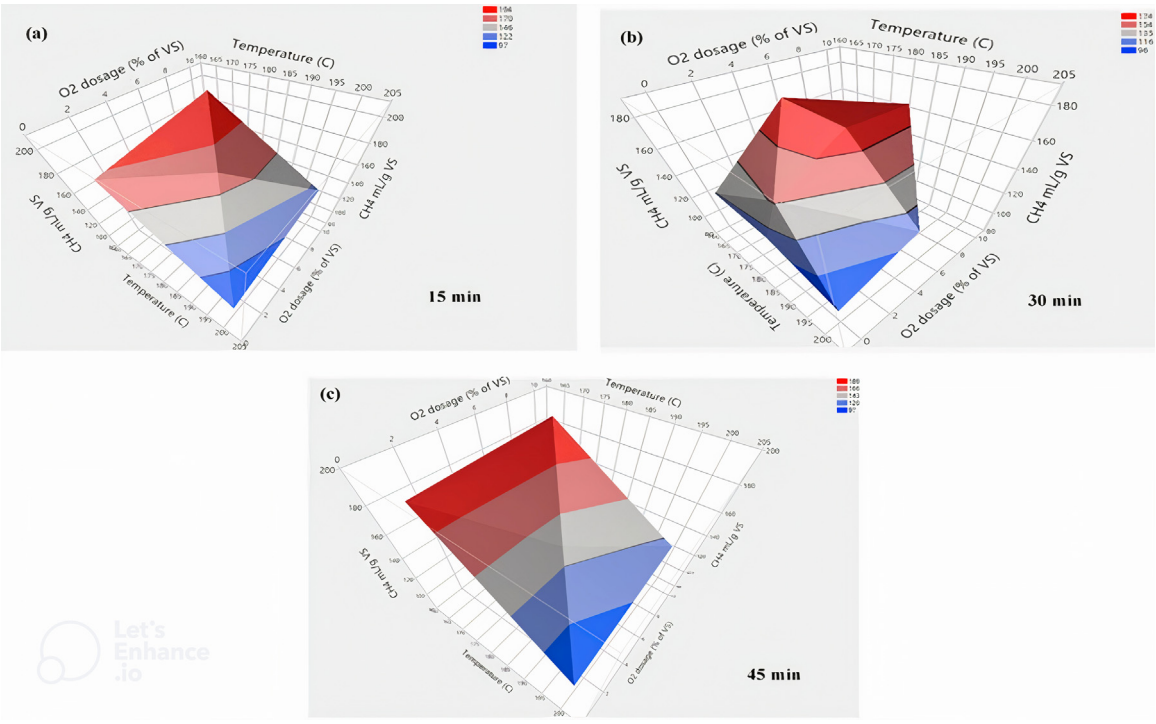


Fig. 3. Response surface plots at different process conditions based on methane yield (ml methane/gVS).

Table 5
Influence the operating factors of AWO pretreatment on methane yield determined by the SAS software.

Operating factors	LogWorth	*P*-value
Temperature	4.804	0.00002
O_2 dosage (%)	2.509	0.00310
Residence time	0.103	0.78845

temperature as the temperature was raised from 100 to 175 °C in semi-continuous flow digesters, but the increases were half the magnitude of those reported by Li and Noike (1992). In a separate study by Wilson et al. (2011), the gas production decreased slightly at 170 °C compared to 150 °C. A previous study reported that AWOEx pretreatment increased the formation of small molecular weight organics, which is easily biodegradable during AD (Hii et al., 2014; Suárez-Iglesias et al., 2017). Further, the radicals produced during AWOEx pretreatment are powerful in oxidizing recalcitrant compounds such as lignin, which might increase the biodegradability of DADSS (Khan and Ahring, 2020).

Previous work has shown that oxygen break-down longer chained molecules into shorter chains, which can be easily degradable during anaerobic digestion (Abelleira-Pereira et al., 2015; Hii et al., 2014; Liang et al., 2020; Lin et al., 2022; M'Arimi et al., 2020; Suárez-Iglesias et al., 2017). It should also be noted that higher O_2 pressure also promoted the breakage of double or triple bonds compounds into single bond, that may also enhance the carbon conversion efficiency into methane during anaerobic digestion and produce more biogas.

Our results indicate that the high temperature enhances the dissolution of DADSS and that O_2 addition helps oxidizing organic compounds in DADSS into digestable molecules.

3.5. Composition analysis of DADSS before and after AWOEx pretreatment

Table 6 shows the influent and effluent concentration of the bioreactors at 182.5 °C, 5.5% O_2, 15 min and 165 °C, 10% O_2, 15 min after pretreatment of DADSS at the optimal AWOEx conditions compared to the control bioreactor. During AD in the control reactor, 9.6%–12% of the VS was degraded, while the reduction of VS was 29.6 to 47% after AD of pretreated material at 165 °C, 30 min and 5.5% O_2 and 182.5 °C, 30 min and 5.5% O_2.The protein content in the effluent sets of the bioreactors 182.5 °C, 15 min and 5.5% O_2 and 182.5 °C, 30 min and 0% O_2 decreased by 46.4% and 29.5%, respectively compared to 9.6% protein content reduction in the control reactor. Since protein constitutes a significant fraction of the organics of the wastewater, its conversion will be reflected in the reduction of volatile solids. The polymers composed of

Table 6
Characteristics of influents and effluents of the bioreactors.

Characteristics	Influent				
	Control reactor	182.5 °C; 15 min; 5.5%	182.5 °C; 30 min; 0%	165 °C; 15 min; 10%	165 °C; 30 min; 0%
Ash (%)	2.92	2.20	2.28	2.36	2.80
Calories (Kcal/100 g)	15	52	26	34	27
Carbohydrates (%)	2.25	2.81	2.70	2.95	2.85
Crude fat (%)	1.69	1.61	< 1.3	< 1.3	< 1.3
Protein (%)	3.01	4.68	4.57	5.41	4.90
Characteristics	Effluent				
Ash (%)	1.71	3.68	2.93	3.25	2.86
Calories (Kcal/100 g)	8	24	15	12	8
Carbohydrates (%)	1.98	0.87	1.78	0.72	1.65
Crude fat (%)	< 1.3	< 1.3	< 1.3	< 1.3	< 1.3
Protein (%)	2.72	1.89	3.22	1.17	2.59

carbohydrate decreased by 69% with a higher conversion rate of hemicellulose (ca.60%) compared to cellulose (ca. 29%) for the bioreactor at 182.5 °C, 15 min and 5.5% O_2. The crude fat content under optimal conditions also reduced in the effluent by 18% compared to 23% reduction in the control for the bioreactor set at 182.5 °C. It should be taken into account that thermal hydrolysis increases the susceptibility of highly unsaturated fatty acids to the reaction of oxygen. For anaerobic digestion to work effectively, polymers in the material must be hydrolyzed to allow for bioconversion to methane gas. Hydrolyzing grease and fat into long chain fatty acids possess a large biogas potential of importance for the methane yield (Wett et al., 2006).

The presence of lipids and proteins in wastewater sludge may generate compounds which are inhibitory to methanogenesis such as ammonia (Liu and Sung, 2002; Tanaka et al., 1997) and long-chained hydrophobic fatty acids (Angelidaki and Ahring, 1992; Chowdhury et al., 2007) through the thermal hydrolysis of proteins and lipids, respectively. Under the conditions giving the optimal methane yield, the biodegradation of the organics significantly improved to 78% in the secondary AD from ca. 50% VS reduction without pretreatment. A previous study by Khan and Ahring (2020) showed enhanced solubilization of carbohydrates after AWOEx pretreatment, which resulted in a higher methane yield during AD of pretreated manure fibers (Khan and Ahring, 2020). In this study, we further saw improved conversion of other components such as proteins and lipids.

The pretreated DADSS after AWOEx at 165 °C with 10% O_2 for 15 min contained higher concentrations of VFAs (1990 mg/L) probably due to the hydrolysis reaction of the acetyl groups of hemicellulose producing acetic acid (Jönsson and Martín, 2016) . Overall, acetic acid was the dominant organic acid in all bioreactors and the bioreactor with the highest VFA concentration further contained the highest concentration of acetic acid (1050 mg/L). The concentration of VFA in the effluents of all bioreactors were, however, low (less than 22 mg/L), indicating that all bioreactors are functioning well and all digestible materials are being fully converted to methane.

4. Conclusions

This study reports for the first-time implementation of Advanced Wet Oxidation & Steam Explosion pretreatment (AWOEx) of dewatered anaerobic digested sewage sludge (DADSS). In this present study, the optimal AWOEx pretreatment conditions for DADSS were identified based on the methane yield. The highest average methane yield was determined to 185 mL/g VS at pretreatment at 165 °C for 15 min residence time and 10% O_2, which was 1.61 times higher than the control reactor (Fig. 1) without pretreatment during AD of the DADSS material. A high methane yield (Fig. 1) was further obtained with the same retention time with a lower oxygen concentration (5.5%), when using a higher temperature (182.5 °C). This clearly show that both oxygen and temperature are important for enhancing the effect of the AWOEx pretreatment on digestibility of sewage sludge during AD as further shown by the response surface model. We further found a strong positive correlation of 0.968 between methane yield and TS/VS ratio. Interestingly, our study clearly demonstrates that not only temperature — but the combination of temperature and oxygen addition plays an important role for enhancing the methane yield during pretreatment. AWOEx pretreatment of DADSS enhanced methane production significantly with over 50% increase compared to standard one-step AD of sewage sludge. This is the report showing that pretreatment of dewatered digested sewage sludge using AWOEx can significantly improve methane production with over 50% increase compared to conventional one-step AD treatment of sewage sludge. Furthermore, we found that the AD process can generate 30% more methane with oxygen added compared to conditions with only heat. Overall, the two-step AD process with pretreatment produced over 90% more methane than the conventional one-step AD leaving only low concentrations of organics behind after the process.

CRediT authorship contribution statement

Nalok Dutta: Conceptualization, Data curation, Writing – original draft, Visualization. **Richard Garrison:** Data curation, Investigation, Conceptualization, Visualization. **Muhammad Usman:** Conceptualization, Data curation, Writing, Visualization. **Birgitte Kiaer Ahring:** Validation, Writing – review & editing, Methodology, Resources, Investigation, Conceptualization, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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